

IMPLEMENTATION OF A VOICE-CONTROLLED SAFETY ASSIST FOR BICYCLISTS

Abstract:

Abstract— The major hurdle for people who wanted to commute via bicycles is the vulnerability to accidents. While there are many possible factors associated with these, reducing the risk of mishaps would always add a significant amount of motivation to push people into this sustainable habit. This project thus aimed at developing a system to assist the rider while tackling the avoidable factors leading to accidents.

The device placed at the handle of the cycle consisted of a microcontroller connected to several sensors which assess the situation. Certain features of the model were controlled by the voice-input of the bicyclist which would be then processed by machine-learning algorithms. The device was capable of controlling the turn lights based on voice input, detecting and alerting the cyclist to obstacles in the path and detecting crashes in the worst case. Thus, the device functioned as an all-round safety-assist to the rider thereby reducing the proneness to accidents.

Keywords—Bicyclists, Safety, Machine-learning, Voice recognition, crash-detection, obstacle-detection.

Introduction:

According to a report presented by the World Health Organization (WHO) in 2020 [1], around 41000 cyclists were killed annually while commuting via road. The major reasons for these accidents were found to be the improper balance of the cycles, non-availability of turn lights to alert the co-motorists of their next move, and minimum to no protection in the case of a crash. With an increase in consciousness towards health post-COVID, there was a rise in the average number of bicyclists commuting over the road [2]. With an increasing number of bicyclists commuting via the road, the concern towards the safety of the bicyclists became more prevalent. It thus became a very important area to explore with regard to different technologies and with that came a necessity to innovate new products concerning the safety of cyclists. While going through the existing innovations available in the market, several things were noticed. Some of these solutions constricted themselves to a few features, while others were too complex and costly to be mounted on a bicycle.

While exploring possible ways to deal with the problems mentioned and the various technologies that could tackle them, the thought of implementing it using a small microcontroller device with its functioning aided by machine

learning algorithms, specifically deep learning was stumbled upon. Traditionally, to indicate the turn of the cycle, the rider raised his respective hand and signalled the oncoming drivers of the turn he was about to take. This however led to an imbalance of the cycle and if the rider was not quick enough to take the control back, it led to a fall. If a prototype could be designed to head-on with the issue, it would be a very effective solution to the problem.

During the literature survey, a few models tackling the issue of bicyclist safety were looked upon. One such system [3] was a voice-controlled model that governed the turn lights of the bicycle. Voice-controlled lights implied that bicyclists need not raise their hands for turns which decreased the risk of accidents. Apart from the turn lights, a smart jacket was also proposed to increase the safety of bicyclists. An upgraded version of this work [4] theorized the use of google maps to indicate the turn in higher wheeled automobiles. This idea of using a computerized voice to drive the system was unique and solved the major problem of inconsistency in the voice signal that was given as the input to the system. In order to understand the concepts behind these models, the studies done in the past years since 2006 pertaining to deep learning applications in speech recognition [5] were referred. These models employed neural networks that were a part of the machine learning algorithms to recognize and process the speech input. The processed input was then compared with the training data of the model. Further, the existing traditional and machine learning-based methods of speech recognition and voice comparison [6] were touched upon for the goal of delivering an effective model. Accurate and precise voice recognition was always a significant achievement because of the wide variety of applications it had. The application that was dealt with in this paper was related to road-safety.

At times, bicyclists were inattentive and tended not to see the vehicles before them, got in too close and suffered with an accident. Helping maintain their distance and alerting bicyclists to impending danger was a significant feature of the model. While there were multiple sensors that achieved this purpose, Ultrasonic sensors [7] provided to be the most cost-effective and reliable amongst them all. These sensors employed sound waves to detect obstacles in the path. The range of these sensors were set to the minimum safe distance necessary between two vehicles and employed to use. These sensors further employed in autonomous control of automobiles [8] wherein the data sent by the sensors was analysed by artificial intelligence to make decisions. This showed how reliable these sensors were, thus raising the safety quotient.

Despite all the safety measures and attentiveness of the bicyclist, accidents occurred. Detecting these crashes and relaying the information to the emergency services reduced the chances of death. An accidental fall could be easily detected by a model based on triaxial accelerometer [9]. It employed an algorithm to process the data of lateral displacements along the axes collected from the accelerometer and detected the kind of fall that occurred to an elderly. Although this was a different kind of application, similar means could be employed to detect the fall of a cycle. A much more precise work employed an Inertial Management Unit (IMU) sensor [10] to detect the crashes.

This sensor used a combination of accelerometer, gyroscopes and magnetometers for that purpose. The same purpose was further satiated by a telematic e-box [11]. An algorithm then analysed the inertial measurements collected by the e-box and classified the crash based on its severity. The severity of this crash and the location of the crash collected by GPS was then relayed by a GSM module. While one specific model employed an external GSM module [12] for this purpose, another made use of an android application [13] to relay the information via mobile. Machine learning had also been applied to detect the crash in the case of a four-wheeler [14]. This system was rendered effective due to the distinctiveness of the sounds pertaining to a car crash which was however not the same in the case of a bicycle.

Hence, the paper proposed to design an assist system capable of indicating turns activated by voice, detect obstacles and relay the vital information in the event of a crash.

Equipment:

- 1) ESP Eye (Microcontroller)
- 2) Ultrasonic sensors (4)
- 3) GPS/GSM module
- 4) LEDs
- 5) LED strip
- 6) Connecting wires
- 7) Accelerometer based sensor such as IMU sensor
- 8) Power source
- 9) Connecting wire

10) Battery equipment:

- TP4056 5V,1A Battery Charging Module: Rs. 100 (pair)
- LM7805 5V IC, Voltage regulator: Rs. 50 (pair)
- DCJ0202, fem power jack: Rs. 74
- 3.7V to 5V Boost Converter Module: Rs. 245
- 3 Pin SPDT Switch: Rs. 50 (pair)
- 470uF, 25V: Rs. 54 (3)
- 100uF,16V: Rs. 54 (3)
- 220 ohms (present)
- 5mm LED (present)
- 2.54mm Female Header (4 needed): Rs. 130 (10)
- 2.54mm Male Headers: Rs. 70 (10)

(OR)

CentloT - BS001 Lithium Battery Shield: Rs. 586

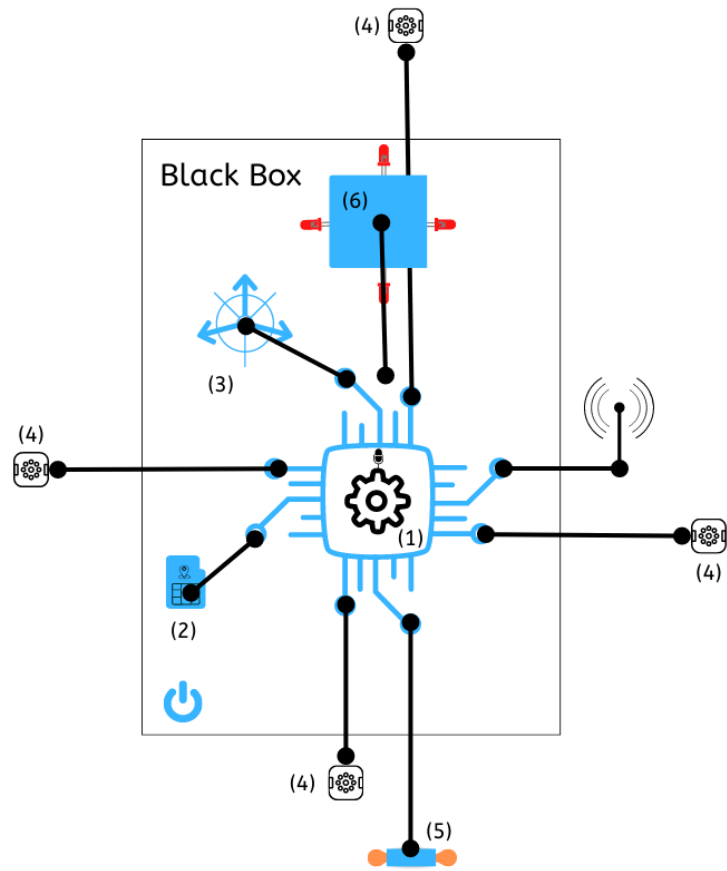
- icall Fully Waterproof Handlebar Mount Holder with 360° Rotation for Motorcycle-Bike-Scooter-Bicycle-Cycle Ideal for Maps and GPS Navigation (Bike Waterproof Mobile Pouch)

Features:

The proposed prototype provided many features to improve bicyclists' safety while on the road. These were as follows:

- 1) All the components were enclosed inside a black box that could easily be mounted on the handle of the bicycle.
- 2) A voice-controlled turn indicator light so that the oncoming vehicles would be notified.
- 3) Alerting the rider to upcoming vehicles from all four directions.
- 4) Detection of falls and crashes that occurred.
- 5) Conveyance of the location and the necessary details to the emergency contact in the event of an accident.

Block Diagram:





Prototype design and Implementation

The proposed Safety assist for bicyclists was designed with the goal of looking over the safety of bicyclists. The block diagram of the system as shown in Fig. 1 gives a brief idea of the components present in the module and the way the system is mounted on the bicycle was shown in Fig. 2. The components corresponding to the numbers that they were labeled with are as follows:

- 1) ESP EYE; 2) GPS/GSM Module; 3) Accelerometer; 4) Ultrasonic sensors;
- 5) Turn-indicator lights; 6) LED array

The design details, specifications and the functions of the modules selected were discussed in the following sub-sections.

- 1) ESP EYE with inbuilt microphone

ESP EYE is a microcontroller based on the ESP32 chip developed by the espressif systems with support for Artificial Intelligence and Machine Learning frameworks. It is a development tool that could be specifically used for Image-processing and Voice-recognition. The essential features include an 8 MegaByte Pseudo-Static RAM, 4 MegaByte Flash memory with connectivity via Wi-Fi and Bluetooth stacks. It additionally has a 2-MegaPixel OV2640 camera, an inbuilt I2S Microphone and an LCD Display. The debugging was performed by a micro-USB port. Additionally, it supports Amazon Web Series providing direct connectivity with IoT services provided by the cloud. ESP-EYE provides the processing power to our module. Additionally, it takes the speech input from the user, processes it using the algorithms embedded in it, recognises the keyword and necessarily performs the function accordingly.

2) GPS/GSM module

The Global Positioning System (GPS) module traces the location of the user and reads it in the format of longitude and latitude. It could continuously provide the data about the location of a person which could be then sent to the microcontroller to be analyzed.

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